

Task Questions

1. Question 1

An AWS Region is a geographical area where AWS data centers are located. Each Region consists of multiple Availability Zones, Availability Zones are separate areas built to stay safe even if something goes wrong in one of them. They are related because they provide options for redundancy and fault tolerance.

2. Question 1

There three types of Elastic Load Balancers:
Classic Load Balancer, Application Load Balancer, and Network Load Balancer.

We should use an Application Load Balancer when we need to route HTTP and HTTPS traffic based on content of the request (such as path or host), or when we need to route requests to multiple services or containers running on a single EC2 instance.

3. Question 1

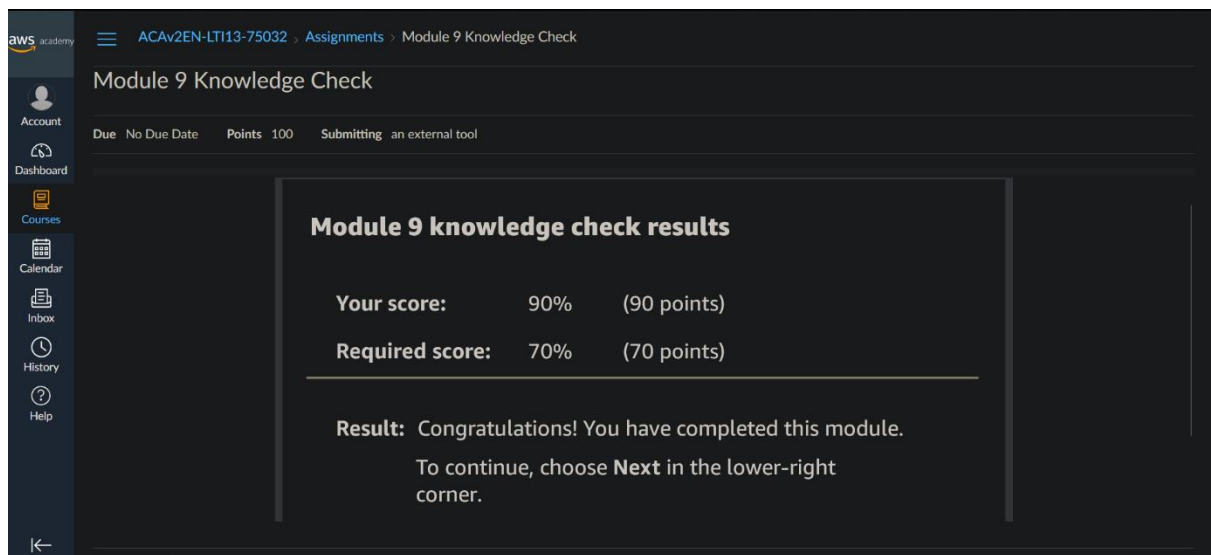
To make an RDS highly available, we can configure it for Multi-AZ (Availability Zone) deployment. This means that a standby replica of the database is automatically created in a different Availability Zone. If the primary database fails, Amazon RDS automatically fails over to the standby replica to minimize downtime.

Quiz

1. Knowledge Check for Module 9

This knowledge check was comprised of the most important questions throughout the module 9. The questions asked was kind of a summary to the whole module 9. In this knowledge check I was able to see my understanding of this module, and the from the mistakes I was able to correct myself of the parts that I was not sure about.

This knowledge check was mainly about Scaling and autoscaling. First question itself was about the EC2 Auto scaling, which statement accurately describes it. There were questions about horizontal scaling and vertical scaling. How autoscaling works in databases. Then there were questions about load balancers. Finally there were questions about high availability, most of the questions were scenario based ones



The screenshot shows the AWS Academy interface for the 'Module 9 Knowledge Check'. The top navigation bar includes the AWS Academy logo, a hamburger menu, the course ID 'ACAv2EN-LTI13-75032', and links to 'Assignments' and 'Module 9 Knowledge Check'. The left sidebar contains navigation icons for Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area displays the 'Module 9 Knowledge Check' title and a table with the following data:

Due	No Due Date	Points	100	Submitting	an external tool
Module 9 knowledge check results					
Your score:	90%	(90 points)			
Required score:	70%	(70 points)			
Result: Congratulations! You have completed this module.					
To continue, choose Next in the lower-right corner.					

Module 9 Guided Lab - Creating a Highly Available Environment

Task 1: Inspecting your VPC

In this task, I explore the configuration of a Virtual Private Cloud (VPC) set up via AWS CloudFormation. Starting in the AWS Management Console, I filter the view to show only the Lab VPC resources. I then review the VPC details, noting its IPv4 CIDR block of 10.0.0.0/16. I examine "Public Subnet 1" within the VPC, observing its IPv4 CIDR block of 10.0.0.0/24 and its associated Availability Zone. Lastly, I review the "Inventory-DB" security group, which currently allows MySQL/Aurora traffic on port 3306 from anywhere within the VPC, noting that this will be later restricted to specific application servers. This inspection gives me a comprehensive understanding of the VPC's structure and traffic management.

The screenshot shows the AWS Management Console interface. The left sidebar contains navigation links for VPC dashboard, EC2 Global View, and various VPC resources. The main content area displays the 'Your VPCs (1/1)' list, filtered by VPC ID 'vpc-02e8a8dafefbadfc70'. The details for this VPC are shown, including its name 'Lab VPC', state 'Available', IPv4 CIDR '10.0.0.0/16', and DHCP options 'dopt-0055'. The 'Details' tab is selected, showing various configuration parameters like DNS hostnames, DNS resolution, and IPv6 pool.

Name	VPC ID	State	IPv4 CIDR	IPv6 CIDR	DHCP opti
Lab VPC	vpc-02e8a8dafefbadfc70	Available	10.0.0.0/16	-	dopt-0055

vpc-02e8a8dafefbadfc70 / Lab VPC

Details

Parameter	Value
VPC ID	vpc-02e8a8dafefbadfc70
State	Available
Tenancy	Default
Default VPC	No
Network Address Usage metrics	Disabled
DHCP option set	dopt-00550fb55a061465c
IPv4 CIDR	10.0.0.0/16
Route 53 Resolver DNS Firewall rule groups	-
DNS hostnames	Enabled
DNS resolution	Enabled
Main route table	rtb-0523e35236a087a53
Main network ACL	acl-08a1f9e5f1bb6efd7
IPv6 pool	-
IPv6 CIDR (Network border group)	-
Owner ID	709137684538

Delving into subnet details, I see that local traffic is routed within the VPC, while internet-bound traffic goes through the internet gateway, making the subnet public.

The screenshot shows the AWS Management Console interface, specifically the 'Subnets (4)' page. The left sidebar contains navigation links for VPC dashboard, EC2 Global View, and various VPC resources. The main content area displays a list of subnets filtered by VPC ID 'vpc-02e8a8dafefbadfc70'. The subnets listed are Public Subnet 2, Private Subnet 1, Private Subnet 2, and Public Subnet 1, all with a state of 'Available'. The details for these subnets are shown, including their names, subnet IDs, states, VPCs, and IPv4 CIDR blocks.

Name	Subnet ID	State	VPC	IPv4 CIDR
Public Subnet 2	subnet-083489e599103c01f	Available	vpc-02e8a8dafefbadfc70 Lab VPC	10.0.1.0/24
Private Subnet 1	subnet-085626b50818143f3	Available	vpc-02e8a8dafefbadfc70 Lab VPC	10.0.2.0/23
Private Subnet 2	subnet-0fab55099201aa93b	Available	vpc-02e8a8dafefbadfc70 Lab VPC	10.0.4.0/23
Public Subnet 1	subnet-09bd74b24aefdc8	Available	vpc-02e8a8dafefbadfc70 Lab VPC	10.0.0.0/24

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Module 9 Guided Lab - Creating a Highly Available Environment

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The screenshot shows the AWS Management Console interface. On the left, the 'VPC dashboard' is selected. The main area displays 'Route tables (1/3) info'. A filter is applied for 'VPC: vpc-02e8a8dafebadc70'. The table lists three route tables: 'Public Route Table' (rtb-050d32adb364f0426), a private route table (rtb-0523e35236a087a53), and 'Private Route Table 1' (rtb-0b75dc0dd3b4581a8). The 'Public Route Table' is selected, showing its details. It has two routes: one for destination 0.0.0.0/0 targeting igw-09ce1b6ba57f43c92, and another for 10.0.0.0/16 targeting 'local'. Both routes are 'Active'.

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
Public Route Table	rtb-050d32adb364f0426	2 subnets	-	No	vpc-02e8a8dafebadc70 Lat
-	rtb-0523e35236a087a53	-	-	Yes	vpc-02e8a8dafebadc70 Lat
Private Route Table 1	rtb-0b75dc0dd3b4581a8	2 subnets	-	No	vpc-02e8a8dafebadc70 Lat

Destination	Target	Status	Propagated
0.0.0.0/0	igw-09ce1b6ba57f43c92	Active	No
10.0.0.0/16	local	Active	No

The Network ACLs are set to default, permitting all traffic, but can be restricted further. I confirm the internet gateway named "Lab IG" is attached to the VPC.

The screenshot shows the 'Network ACLs (1/1) info' page. A filter is applied for 'VPC ID: vpc-02e8a8dafebadc70'. The table lists one Network ACL: 'acl-08a1f9e5f1bb6efd7', which is associated with 4 subnets, is the default, and belongs to VPC ID vpc-02e8a8dafebadc70 / Lab VPC. The 'Details' tab is selected, showing the Network ACL ID, Owner (709137684538), and VPC ID.

Name	Network ACL ID	Associated with	Default	VPC ID	Inbo
-	acl-08a1f9e5f1bb6efd7	4 Subnets	Yes	vpc-02e8a8dafebadc70 / Lab VPC	2 Inb

Network ACL ID	Associated with	Default	VPC ID
acl-08a1f9e5f1bb6efd7	4 Subnets	Yes	vpc-02e8a8dafebadc70 / Lab VPC

The screenshot shows the 'Internet gateways (1/1) info' page. A filter is applied for 'VPC ID: vpc-02e8a8dafebadc70'. The table lists one Internet Gateway: 'Lab IG' (igw-09ce1b6ba57f43c92), which is 'Attached' to VPC ID vpc-02e8a8dafebadc70 | Lab VPC. The 'Details' tab is selected, showing the Internet gateway ID, State (Attached), VPC ID, and Owner (709137684538).

Name	Internet gateway ID	State	VPC ID	Owner
Lab IG	igw-09ce1b6ba57f43c92	Attached	vpc-02e8a8dafebadc70 Lab VPC	709137684538

Internet gateway ID	State	VPC ID	Owner
igw-09ce1b6ba57f43c92	Attached	vpc-02e8a8dafebadc70 Lab VPC	709137684538

Lastly, I review the "Inventory-DB" security group, which currently allows MySQL/Aurora traffic on port 3306 from anywhere within the VPC, noting that this will be later restricted to specific application servers. This inspection gives me a comprehensive understanding of the VPC's structure and traffic management.

The screenshot displays the AWS Management Console interface. The left-hand navigation pane shows the 'Security groups' section expanded, with 'Network Firewall' and 'Virtual private network (VPN)' also visible. The main content area is titled 'Security Groups (1/3) Info'. A search bar at the top of the main area contains the VPC ID 'vpc-02e8a8dafefadfc70'. Below the search bar, a table lists the security groups associated with this VPC:

	Name	Security group ID	Security group name	VPC ID	Description
☒	Inventory-DB	sg-080eb0ab44b4c5134	Inventory-DB	vpc-02e8a8dafefadfc70	Enable access to
☐	Inventory-App	sg-05174b72a3d5cad15	Inventory-App	vpc-02e8a8dafefadfc70	Enable access to
☐	-	sg-0803a581d179ad3c5	default	vpc-02e8a8dafefadfc70	default VPC sec

Below the table, the 'Inbound rules (1)' section is shown. It contains a search bar and a table with one rule:

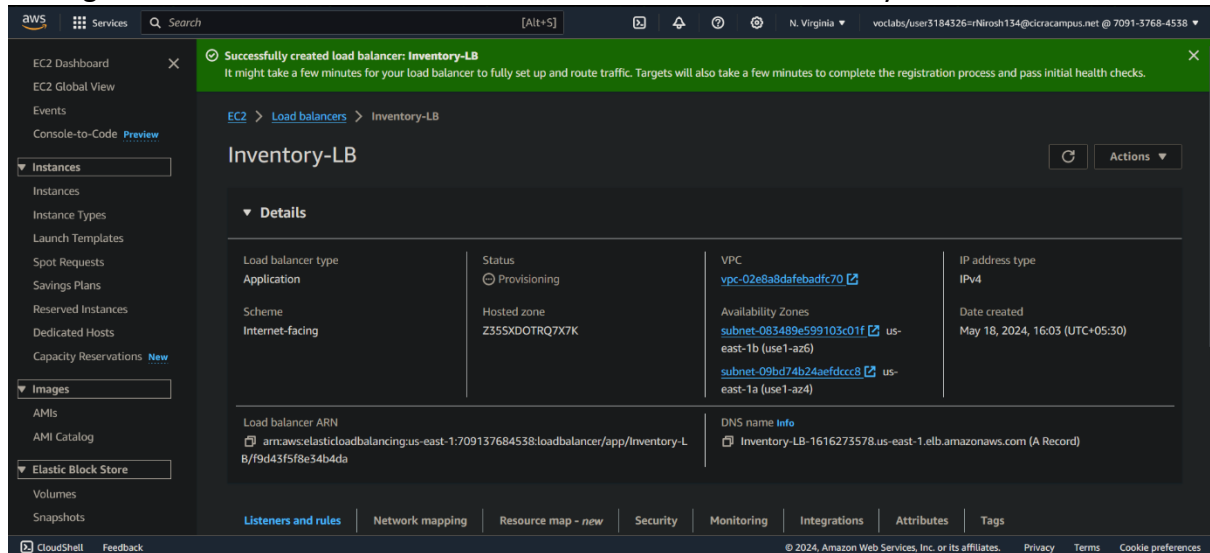
	Name	Security group rule...	IP version	Type	Protocol	Port range
☐	-	sg-09d9460e41bf60f88	IPv4	MySQL/Aurora	TCP	3306

The bottom of the console shows the footer with copyright information: '© 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences'.

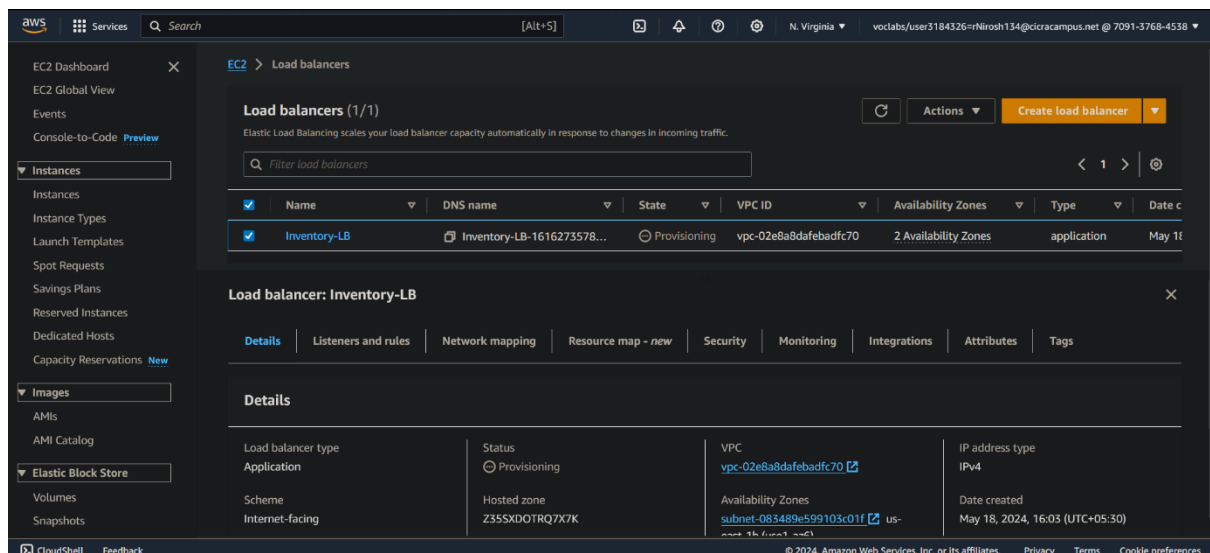
SIT233 – Cloud Computing
Module 9 Guided Lab - Creating a Highly Available
Environment
Task 2: Creating an Application Load Balancer

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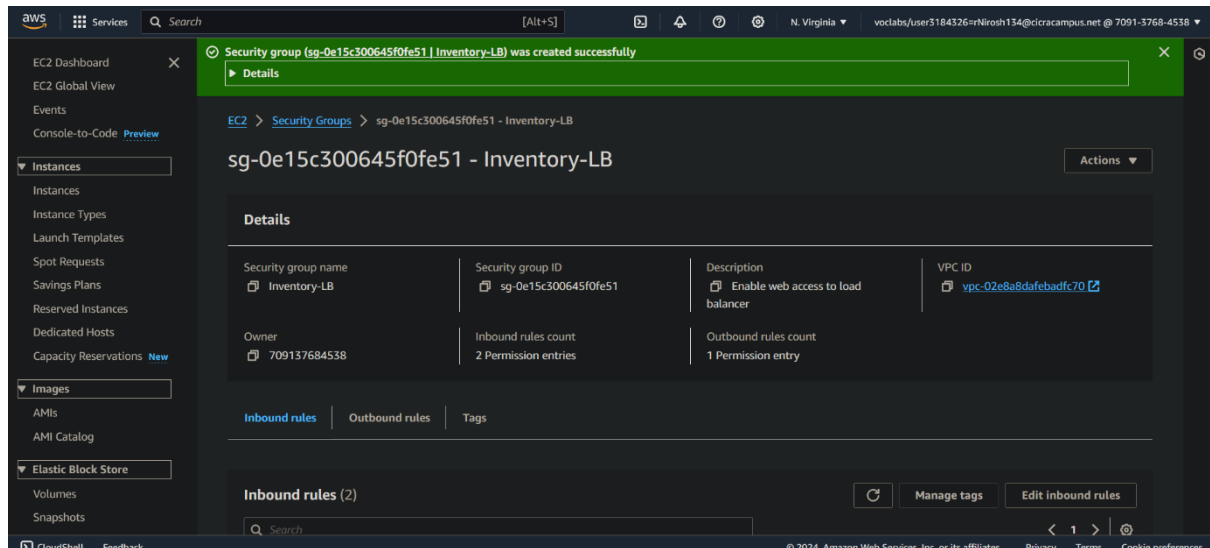
In this task, I set up an Application Load Balancer (ALB) to distribute traffic across multiple servers, ensuring high availability. First, I navigated to the EC2 section in the AWS Management Console and chose to create an ALB named "Inventory-LB".



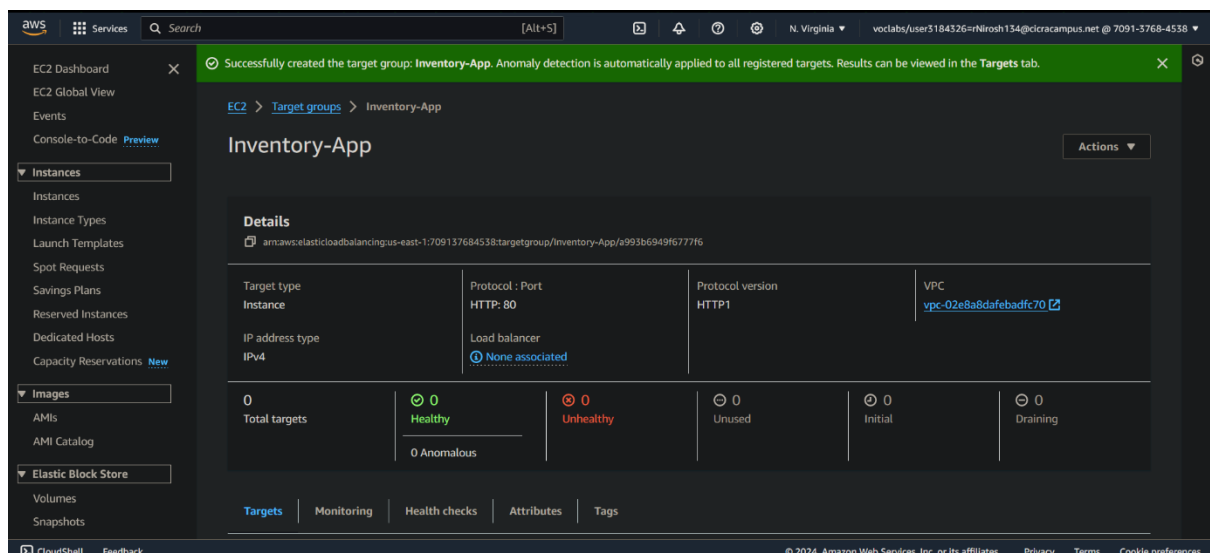
I selected "Lab VPC" and configured it to use both public subnets in two Availability Zones. Next, I created a new security group named "Inventory-LB" to allow HTTP and HTTPS traffic from anywhere and attached it to the load balancer, removing the default security group.



In the Listeners and routing section, I created a target group called "Inventory-App" within the same VPC, adjusting the health check settings for quicker verification. After creating the target group, I set the load balancer to forward HTTP traffic to this group

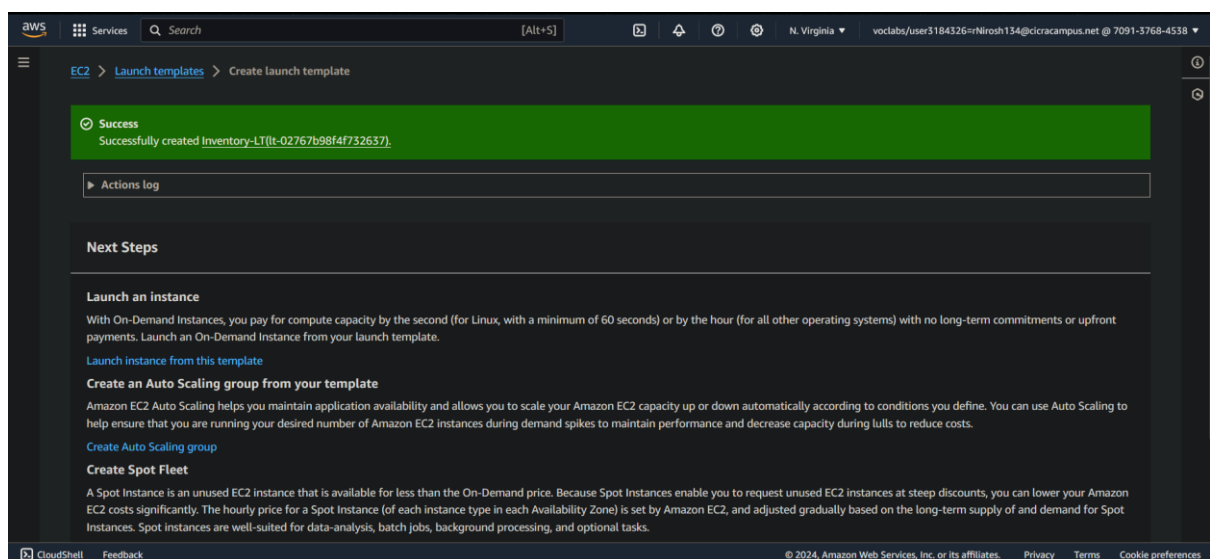
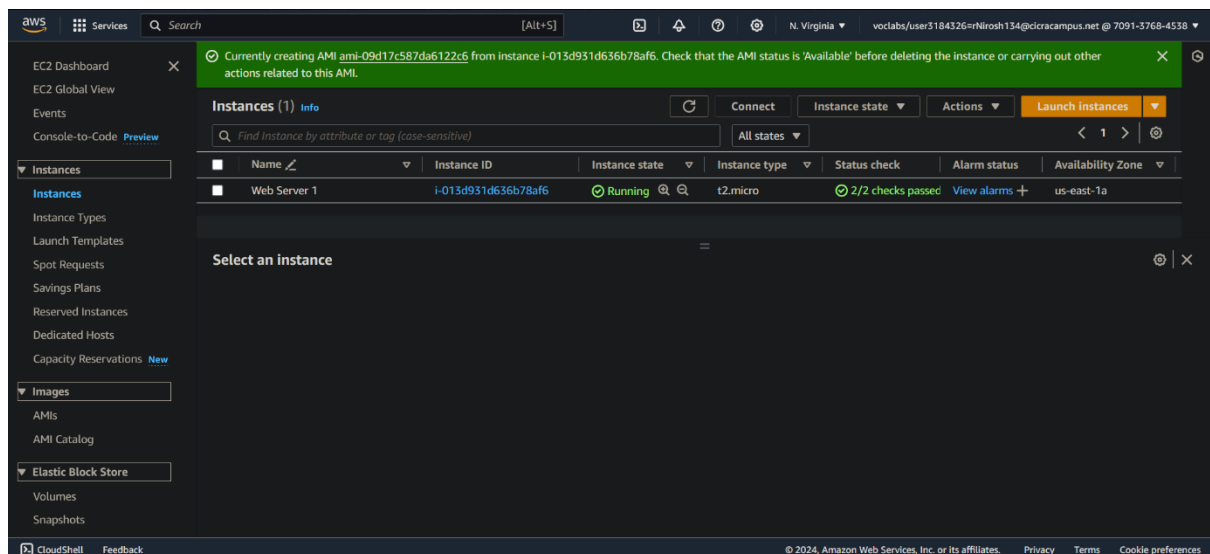


Finally, I created the load balancer and confirmed its successful setup by viewing it in the console. This ensures that traffic is efficiently managed and distributed across multiple servers for high availability.

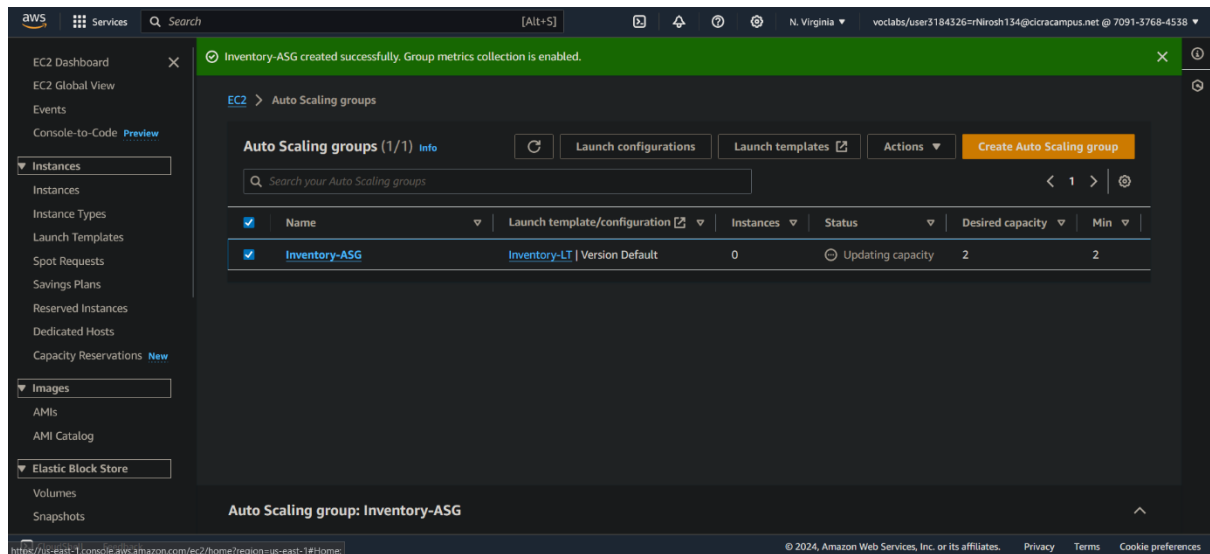


Task 3: Creating an Auto Scaling group

In this task, I set up an Auto Scaling group to ensure my application can handle varying traffic loads by automatically adjusting the number of EC2 instances. I start by creating an Amazon Machine Image (AMI) from an existing web server. Then, I create a launch template named "Inventory-LT" using this AMI, specifying instance details and a startup script. Next, I create an Auto Scaling group called "Inventory-ASG" that uses this template, selecting private subnets in two Availability Zones for better security and availability.

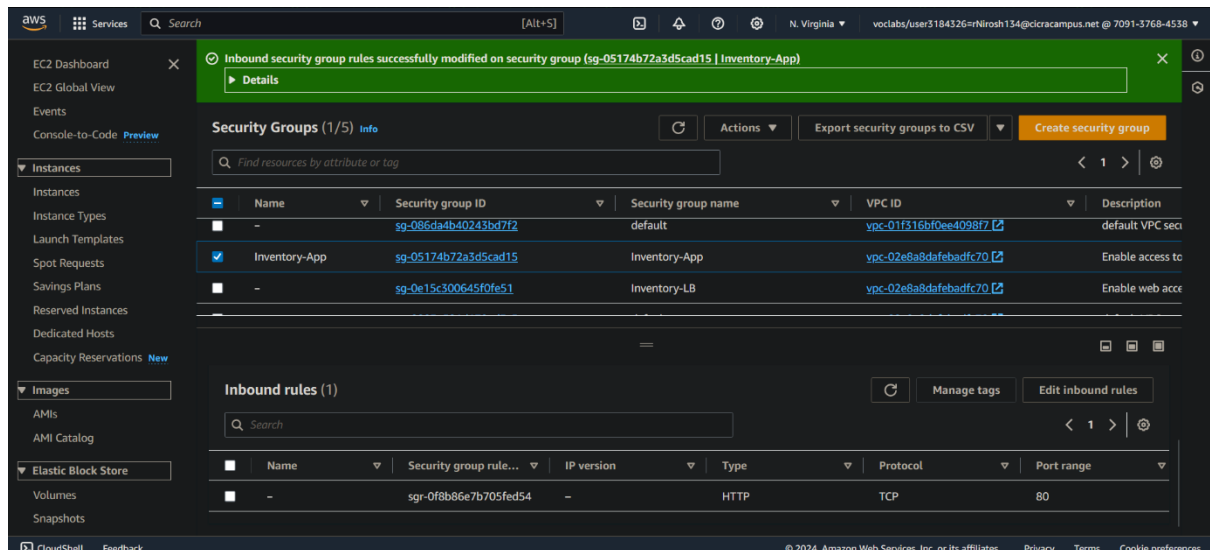


I configure it to always maintain two instances, attach it to the existing load balancer, and enable CloudWatch monitoring for quick scaling responses. Finally, I verify that the Auto Scaling group successfully launches and maintains the desired number of instances.

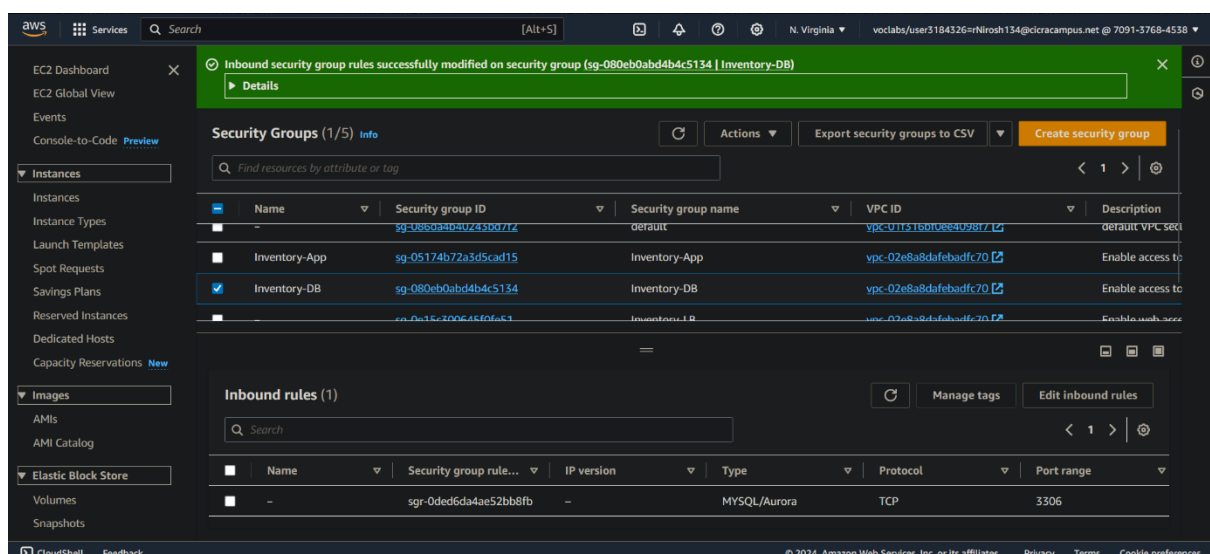


Task 4: Updating security groups

In this task, I set up security groups to enforce a three-tier architecture for my application. First, I confirmed the load balancer security group is already configured to accept all incoming HTTP and HTTPS traffic. Then, I updated the application security group ("Inventory-App") to only accept HTTP traffic from the load balancer.



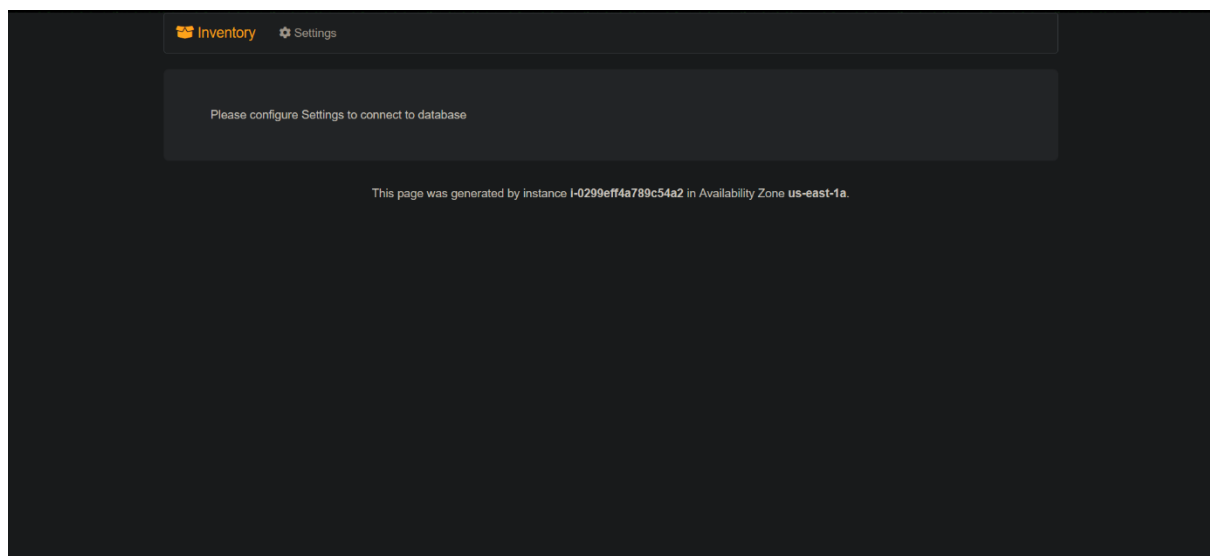
Finally, I configured the database security group ("Inventory-DB") to accept MySQL traffic only from the application servers. This setup ensures that each tier only communicates with the appropriate layer, enhancing the security of my application.



Task 5: Testing the application

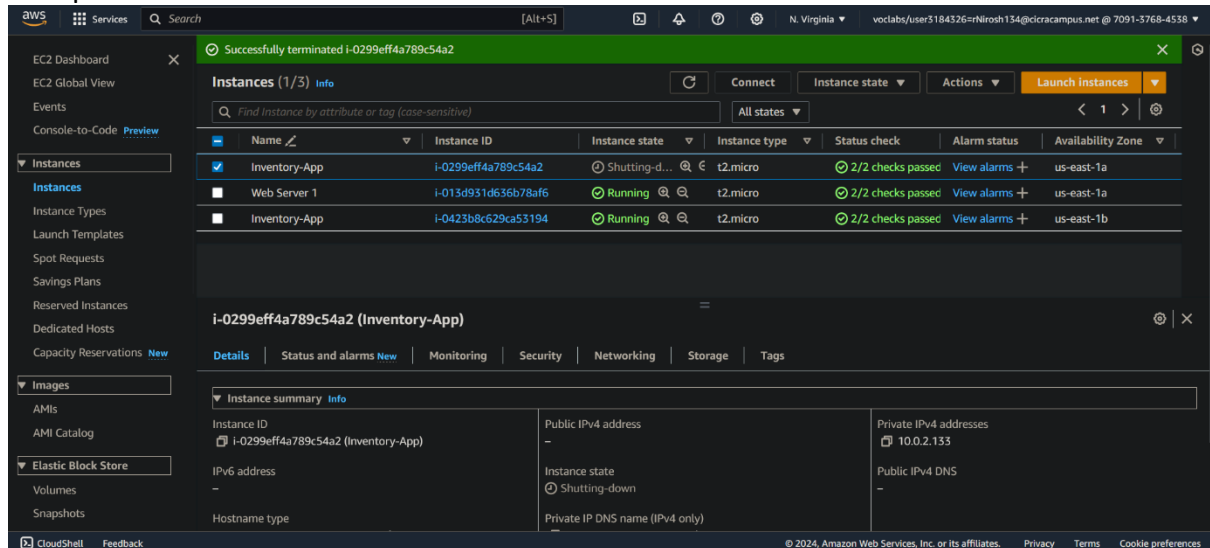
In this task, I'm testing my application to confirm it's running and highly available. First, I navigate to the "Target Groups" section and select "Inventory-App." I check the "Targets" tab to ensure both instances show a healthy status, refreshing the page as needed. Next, I retrieve the DNS name of my load balancer from the "Load Balancers" section, copying it to my clipboard.

I open a new browser tab, paste the DNS name, and press ENTER to access the application. The load balancer directs my request to one of the EC2 instances, and I can see the instance ID and Availability Zone at the bottom of the webpage. Reloading the page toggles between the two instances, confirming that the load balancer distributes traffic across them.

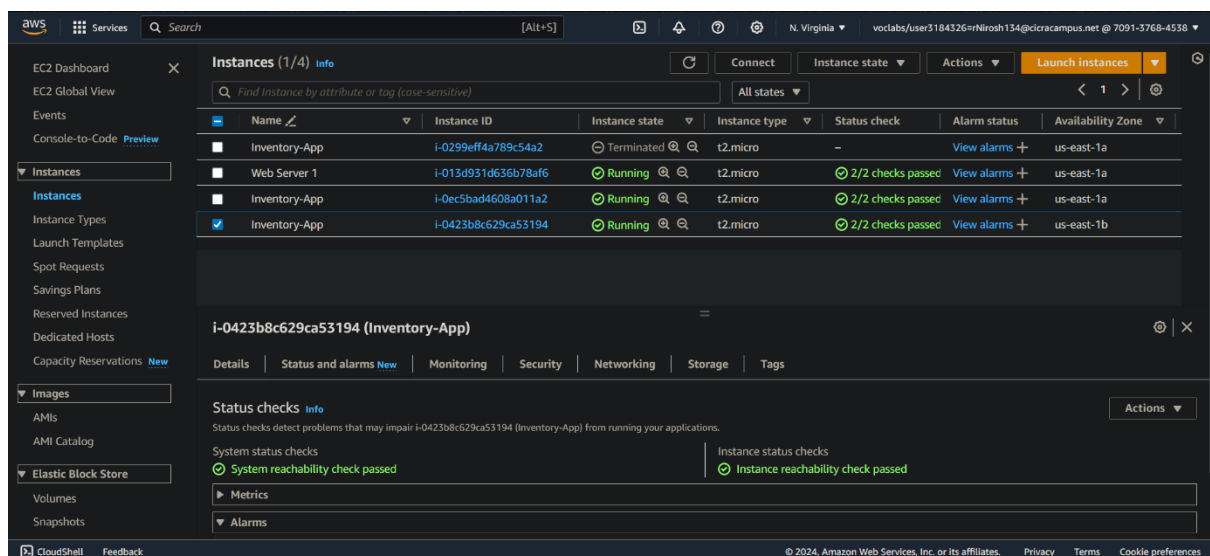


Task 6: Testing high availability

In this task, I'm testing the high availability of my application by pretending one of the EC2 instances fails. I go to the EC2 console, choose an instance, and terminate it. Afterward, I observe that the load balancer automatically directs all traffic to the remaining instance. Then, I refresh my web application page and notice that it stays available without any disruption.



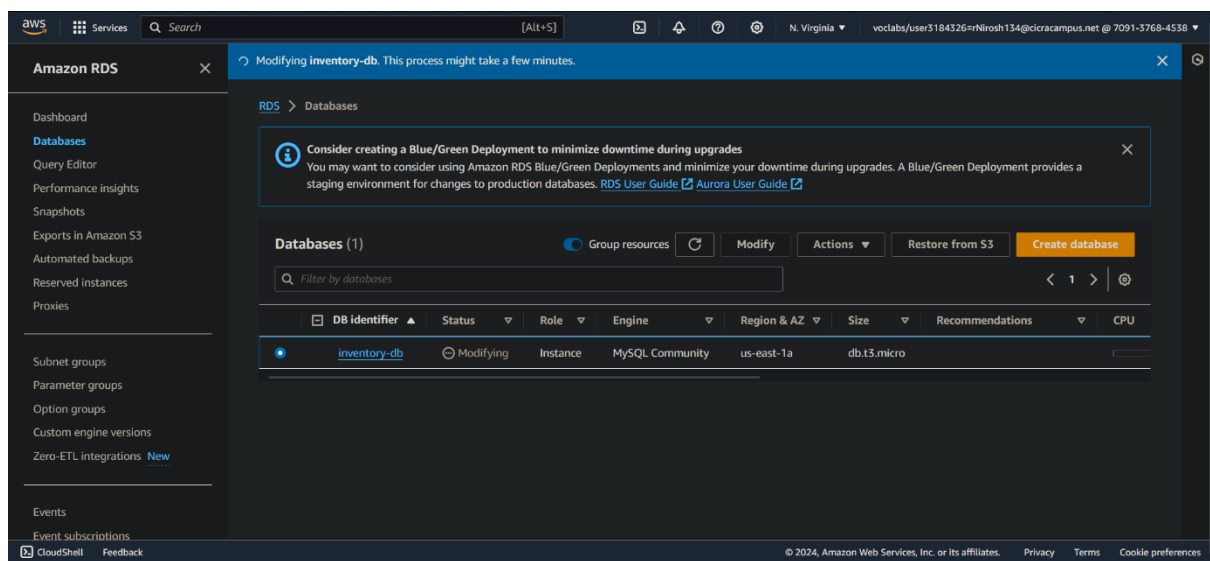
Eventually, a new instance is automatically launched by the Auto Scaling group to replace the terminated one. I keep an eye on the EC2 console until I see the new instance appear, and once it passes its health checks, the load balancer starts distributing traffic evenly between both instances again. This test proves that my application can handle failures and remains highly available.



Optional task 1: Making the database highly available

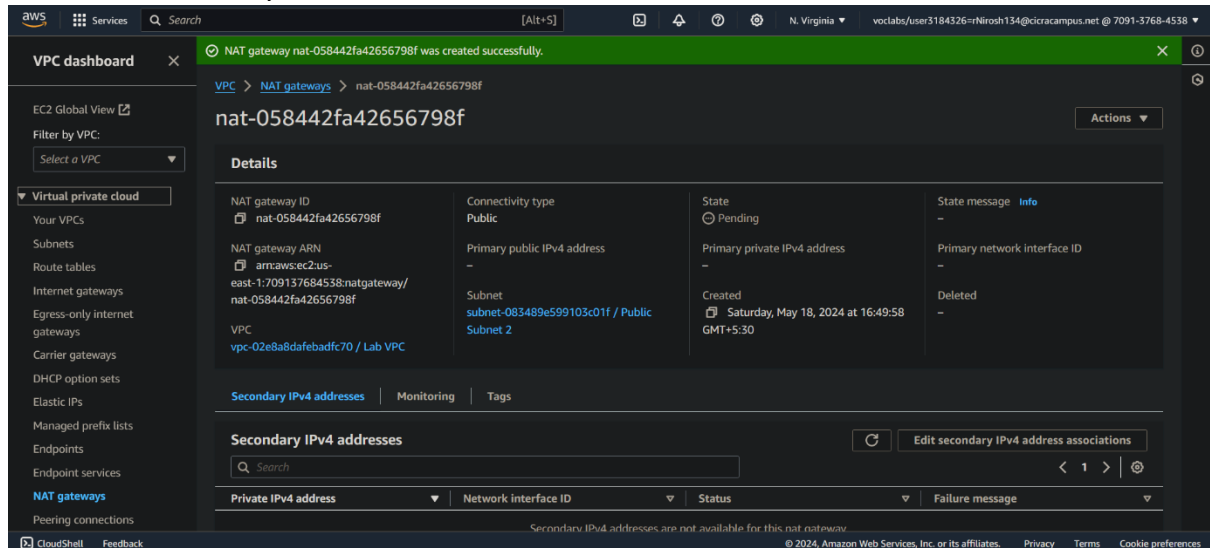
In this optional task, I'm improving the reliability of the Amazon RDS database by setting it up to run across multiple Availability Zones. I access the RDS service, select my database instance, and choose to modify its configuration.

By enabling Multi-AZ deployment, the database operates across multiple data centers, ensuring continuous availability. I also scale up the database instance to increase its size and storage capacity. Once I apply these changes immediately, the database undergoes modification without any need for me to wait for completion.

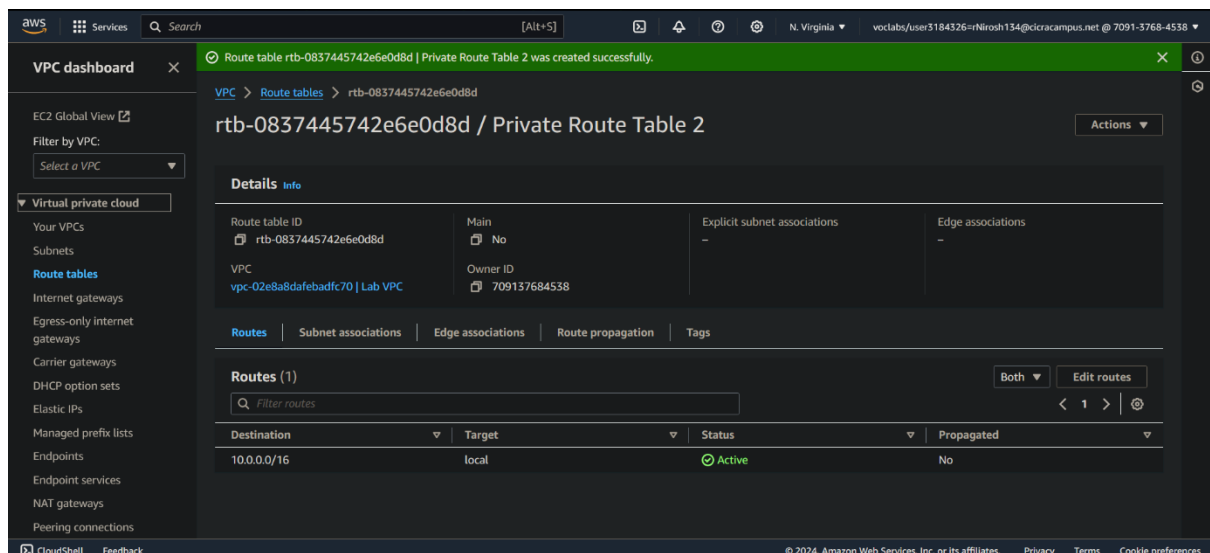


Optional task 2: Configuring a highly available NAT gateway

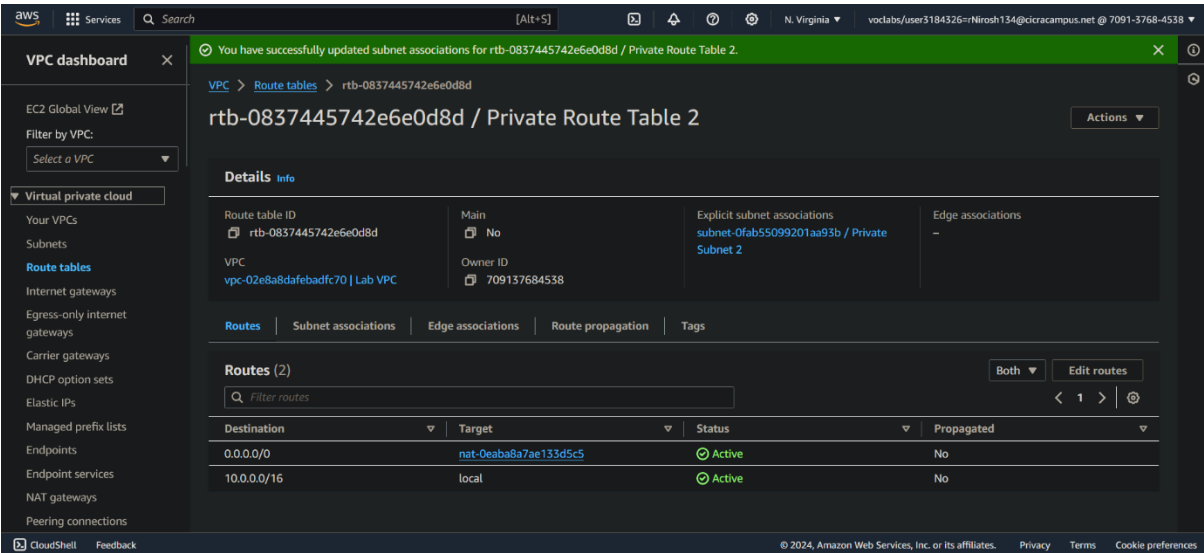
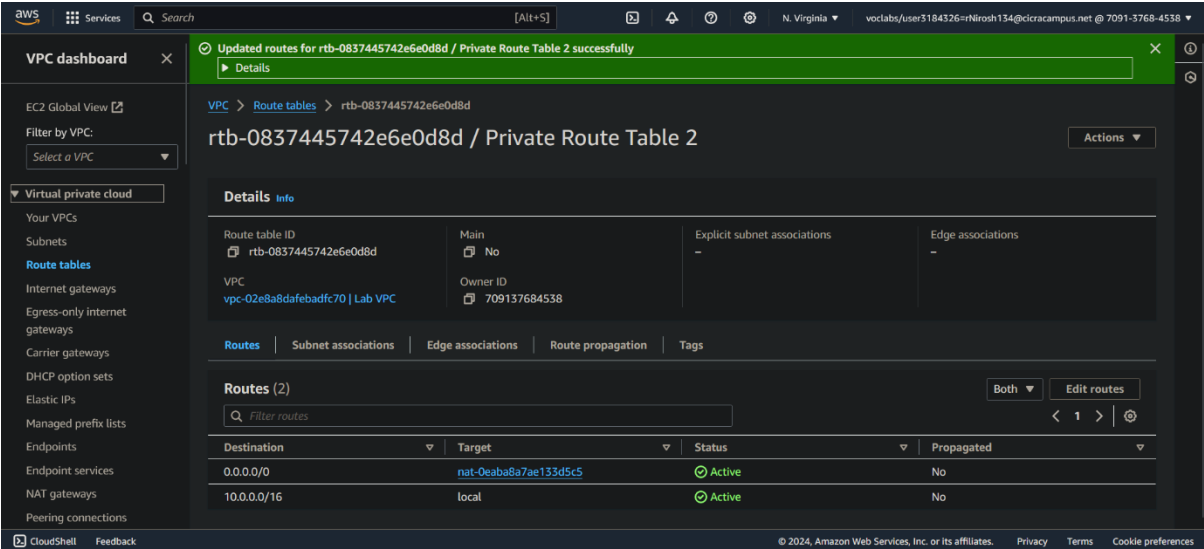
In this optional task, I'm improving the reliability of internet access for the application servers, which currently rely on a single NAT gateway. If that gateway fails, the servers lose internet connectivity



To fix this, I create another NAT gateway in a different Availability Zone. This ensures that even if one zone goes down, the servers can still access the internet through the backup gateway.



After setting up the new gateway and adjusting the route tables, both gateways work together to keep internet access available at all times, making the system highly resilient.



SIT233 – Cloud Computing
Module 9 Guided Lab - Creating a Highly Available
Environment

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Grade

My grade is 75/75

The screenshot shows the AWS Academy interface for an assignment. The left sidebar contains navigation links: Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area is titled 'Module 9 Guided Lab - Creating a Highly Available Environment' and includes a 'Submit' button and tabs for 'Details', 'AWS', 'Start Lab', 'End Lab', '1:38', 'Instructions', 'Grades', and 'Actions'. The 'Details' tab is active, showing a 'Lab overview and objectives' section with the text: 'Critical business systems should be deployed as *highly available applications*—that is, applications remain operational even when some components fail. To'. A terminal window is open, showing a bash prompt and the command 'tee_u_3866282@runweb124750:~\$'. The 'Grades' tab is also visible, showing a 'Total score' of 75/75 and a list of tasks: '[Task 2A] Inventory-LB was created', '[Task 2B] Inventory-LB is an ALB', '[Task 2C] Inventory-LB is in Lab VPC', '[Task 2D] ALB is using Public Subnet 1', and '[Task 2E] ALB is using Public Subnet 2'. The right sidebar shows the 'Submission' status: 'May 18 at 8:46pm', 'Submission Details', 'Grade: 100 (100 pts possible)', 'Graded Anonymously: no', and 'Comments: No Comments'.